



Palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols

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ABSTRACT

A palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols to access aryl benzoate derivatives is described. The procedure tolerates a series of functional groups, such as methoxy, nitro, cyano, chloro, fluoro and bromo groups. Thus, it represents a practically alternative method to access aryl benzoate derivatives.

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1. Introduction

Esterification is one of the most fundamental and important reactions in organic synthesis.¹ Although several methods have been exploited and developed, the search for new environmentally friendly, atom-efficient methods, which avoid the use of large amounts of condensing reagents and activators have attracted increasing interest.² From the atom-economic point of view, the oxidative couplings from aldehydes to access esters have received much attention.³ However, long-term storage and use of aldehydes may be troublesome due to their inherent reactivity. An attractive alternative is the direct catalytic oxidation of alcohols to esters.⁴ Corey detailed the two-step oxidation of allylic alcohols to esters in the presence of sodium cyanide and MnO₂.⁵ Chromium⁶ and iodine⁷ have also been reported to oxidize alcohols to esters. However, a large excess or stoichiometric amount of a toxic oxidant was required in the above transformations. In 1987, Murahashi described a ruthenium-catalyzed oxidative transformation of alcohols and aldehydes to esters and lactones.⁸ In 2002, Hiroi developed an oxidative lactonization of 1,4- or 1,5-diols catalyzed by a novel Ir catalyst.⁹ Subsequently, Hartwig and Ikariya reported ruthenium-catalyzed dehydrogenative cyclization of 1,4-diols to lactones, respectively.¹⁰ Milstein designed a novel ruthenium complexes to catalyze dehydrogenation of primary alcohols to esters or acetals using a harsh condition (>150 °C).¹¹ Recently, Scheidt demonstrated a tandem oxidation of alcohols to esters using *N*-heterocyclic carbenes as catalysts,¹² which required superstoichiometric amounts of MnO₂ as the oxidant. Thus, to develop a facile and versatile procedure on such transformation still remains a highly desired goal for organic chemists. Moreover, less

attention has been paid to the synthesis of aryl benzoate derivatives despite their importance as building blocks of numerous active compounds.^{13,14} Very recently, we developed a palladium/NHC-catalyzed oxidative esterification of aldehydes with phenols.¹⁵ Lei and Beller developed palladium-catalyzed aerobic oxidative esterification of benzylic alcohols, respectively.¹⁶ Herein, we report a palladium/NHC-catalyzed tandem oxidation/aromatic esterification of benzylic alcohols with phenols to access aryl benzoate derivatives, using O₂ as the clean terminal oxidant.

2. Result and discussion

Initial studies were performed by using the reaction of phenylmethanol with 4-nitrophenol under oxygen as the model reaction, employing Pd(OAc)₂ as the catalyst (Table 1). Only a trace of the product was formed in the absence of ligand as determined by GC–MS (Table 1, entry 1). Various precursors of *N*-heterocyclic carbene (NHC) (Fig. 1) were investigated for this reaction. To our delight, an excellent yield was obtained with **L5** (Table 1, entry 6). When the reaction was carried out under nitrogen, no product was detected by GC–MS (Table 1, entry 8). This result suggested that the oxygen played a significant role in the reaction. The choice of solvent was also crucial to the reaction and xylene appeared to be the best among the solvents (Table 1, entries 6 and 9–11). Moreover, the base used also had an important effect in the reaction and Na₂CO₃ proved to be the best. In addition, only a trace of the product was detected by GC–MS in the absence of Pd(OAc)₂ (Table 1, entry 6). Conducting the reaction on a 2 mmol scale provided **3aa** in an acceptable 89% yield.

Encouraged by these results, we further pursued the scope of the process with respect to the phenols. As expected, the reaction proceeded smoothly with yields ranging from good to excellent and tolerated various functional groups, such as methoxy, nitro, cyano,

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Table 1
Selected results of screening the optimal conditions

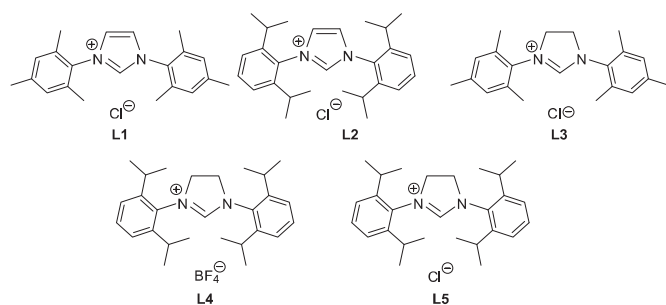
Entry	Ligand	Base	Solvent	Yield ^a (%)
1	—	Na ₂ CO ₃	Xylene	Trace
2	L1	Na ₂ CO ₃	Xylene	10
3	L2	Na ₂ CO ₃	Xylene	16
4	L3	Na ₂ CO ₃	Xylene	65
5	L4	Na ₂ CO ₃	Xylene	82
6	L5	Na ₂ CO ₃	Xylene	95
7	L5	Na ₂ CO ₃	Xylene	Trace ^b
8	L5	Na ₂ CO ₃	Xylene	81 ^c
9	L5	Na ₂ CO ₃	Xylene	<5 ^d
10	L5	Na ₂ CO ₃	DMF	<5
11	L5	Na ₂ CO ₃	Toluene	43
12	L5	Na ₂ CO ₃	1,4-Dioxane	30
13	L5	K ₂ CO ₃	Xylene	61
14	L5	Cs ₂ CO ₃	Xylene	<5
15	L5	NaHCO ₃	Xylene	53
16	L5	<i>t</i> -BuOK	Xylene	43
17	L5	—	Xylene	41

^a All reactions were run with 4-nitrophenol **2a** (0.2 mmol), phenylmethanol **1a** (0.3 mmol), base (0.1 mmol, 0.5 equiv), Pd(OAc)₂ (5 mol %) and ligand (10 mol %) in dry solvent (2 mL) under O₂ at 130 °C for 36 h. Isolated yields.

^b In the absence of Pd(OAc)₂.

^c Under air.

^d Under N₂.

**Fig. 1.** Selected imidazolium salts screened.

chloro, fluoro and bromo groups. Particularly, halogen-substituted phenols worked well, albeit a lower yield was obtained for bromine (Table 2, entries 2–5). The surviving halogen atom (X=Cl, Br) could be valuable for further manipulation. Generally, electron-deficient phenols reacted with phenylmethanol easily and gave the products in higher yields than electron-rich phenols (e.g., **2a**, **2b**, **2c** and **2d** vs **2h**, **2i**, **2j** and **2k**). Steric hindrance on phenols with electron-withdrawing groups had no obvious effect on the reaction (Table 2, entries 3 and 4). Nevertheless, the hindrance on phenols had a sharp effect on the reaction for those electron-rich analogues (Table 2, entries 9–11). For example, **3ak** was obtained in only 35% yield when **2k** was subjected to the procedure.

Next, we turned our attention to the scope of benzylic alcohols, and the results are shown in Table 3. Once again, the reaction proceeded smoothly with moderate to good yields and tolerated various functional groups, such as nitro, cyano, chloro, fluoro, trifluoromethyl and methoxycarbonyl groups. Generally, the benzylic alcohols possessing electron-withdrawing groups gave slightly higher yields than those of electron-donating groups (Table 3, entries 9–14). It is worth noting that a saturated ester was obtained when allyl alcohol was subjected to the procedure (Table 3, entry 14).¹⁷

Table 2
Palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of phenylmethanol with phenols

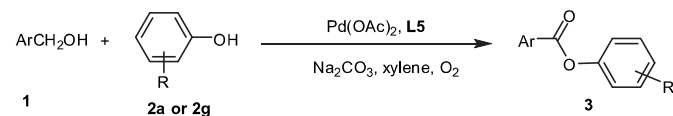
Entry	Phenol 2	Product 3	Yield ^a (%)
1	4-nitrophenol 2a	4-nitrobenzoate 3aa	95
2	4-fluorophenol 2b	4-fluorobenzoate 3ab	93
3	4-chlorophenol 2c	4-chlorobenzoate 3ac	91
4	2-chlorophenol 2d	2-chlorobenzoate 3ad	89
5	4-bromophenol 2e	4-bromobenzoate 3ae	60
6	4-cyanophenol 2f	4-cyanobenzoate 3af	70
7	phenol 2g	benzoate 3ag	75
8	4-methylphenol 2h	4-methylbenzoate 3ah	61
9	4-methoxyphenol 2i	4-methoxybenzoate 3ai	64
10	3-methoxyphenol 2j	3-methoxybenzoate 3aj	82
11	2-methoxyphenol 2k	2-methoxybenzoate 3ak	35
12	1-naphthol 2l	1-naphthoate 3al	88

^a Reaction conditions: **1a** (0.3 mmol), **2** (0.2 mmol), Pd(OAc)₂ (5 mol %), **L5** (10 mol %) and Na₂CO₃ (0.1 mmol) in dry xylene (2 mL) under O₂ at 130 °C for 36 h. Isolated yield.

More experiments were carried out to gain preliminary insight into the reaction mechanism. Conversion of benzylic alcohol to the aromatic aldehyde was observed under the standard procedure by GC–MS. Moreover, when naphthalen-1-ylmethanol **1f** was subjected to the standard procedure without **L5**, 1-naphthaldehyde was isolated in 70% yield (Scheme 1).¹⁸ These results could disclose the possibility of a tandem oxidation of benzylic alcohol to aromatic aldehyde followed by an esterification pathway. However, the mechanism in detail kept unclear in current stage.

Table 3

Palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols

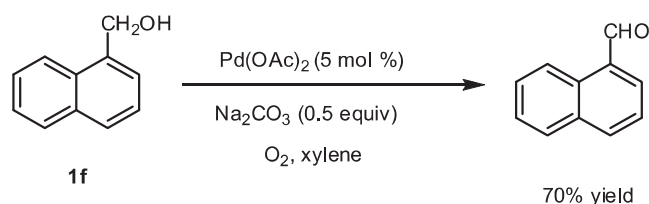


Entry	Substrate 1	Product 3	Yield ^a (%)
1			82
2			52
3			80
4			60
5			82
6			83
7			78
8	1e		46
9			80
10			82
11			76
12			65
13			61
14			67

^a Reaction conditions: **1** (0.3 mmol), **2** (0.2 mmol), Pd(OAc)₂ (5 mol %), **L5** (10 mol %) and Na₂CO₃ (0.1 mmol) in dry xylene (2 mL) under O₂ at 130 °C for 36 h. Isolated yield.

3. Conclusions

In conclusion, we have developed a palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic



Scheme 1. Pd(OAc)₂-catalyzed oxidation of naphthalen-1-ylmethanol to 1-naphthaldehyde by molecular oxygen.

alcohols with phenols to access aryl benzoate derivatives. It obviates the employment of cyanide and super-stoichiometric amounts of MnO₂. This operationally simple one-pot process uses O₂ as the clean oxidant, producing esters in good to excellent yields.

4. Experimental section

4.1. General procedure

¹H NMR and ¹³C NMR spectra were measured on a 500 MHz Bruker spectrometer (¹H 500 MHz, ¹³C 125 MHz), using CDCl₃ as the solvent with tetramethylsilane (TMS) as the internal standard at room temperature. Chemical shifts (δ) are given in parts per million relative to TMS, the coupling constants *J* are given in hertz. Column chromatography was performed using EM Silica gel 60 (300–400 mesh).

4.2. Typical experimental procedure of palladium/NHC-catalyzed tandem benzylic oxidation/oxidative esterification of benzylic alcohols with phenols

Under oxygen, a reaction tube was charged with phenols (0.2 mmol), benzylic alcohols (0.3 mmol), Pd(OAc)₂ (2.2 mg, 5 mol %), **L5** (8.5 mg, 10 mol %), Na₂CO₃ (10.6 mg, 0.1 mmol) in dry xylene (2 mL). After the mixture was stirred for 36 h at 130 °C, the solvent was evaporated under reduced pressure and the residue was purified by flash column chromatography on silica gel to give the product.

4.3. Analytical data of products

4.3.1. 4-Nitrophenyl benzoate (3aa)¹⁹. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.36 (d, *J*=9.0 Hz, 2H), 8.24–8.22 (m, 2H), 7.71 (t, *J*=7.5 Hz, 1H), 7.59–7.56 (m, 2H), 7.45 (d, *J*=9.0 Hz, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.2, 155.7, 145.4, 134.2, 130.3, 128.8, 128.6, 125.3, 122.6.

4.3.2. 4-Fluorophenyl benzoate (3ab)^{14h}. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.19 (d, *J*=7.7 Hz, 2H), 7.64 (t, *J*=7.4 Hz, 1H), 7.51 (t, *J*=7.4 Hz, 2H), 7.19–7.17 (m, 2H), 7.12–7.09 (m, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.2, 160.3 (d, *J*_{C–F}=242.8 Hz), 146.8 (d, *J*_{C–F}=2.8 Hz), 133.7, 130.2, 129.3, 128.6, 123.1 (d, *J*_{C–F}=8.4 Hz), 116.1 (d, *J*_{C–F}=23.4 Hz).

4.3.3. 4-Chlorophenyl benzoate (3ac)²⁰. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.19 (d, *J*=7.5 Hz, 2H), 7.66–7.63 (m, 1H), 7.53–7.50 (m, 2H), 7.40 (d, *J*=8.0 Hz, 2H), 7.17 (d, *J*=8.0 Hz, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.0, 149.4, 133.8, 131.3, 130.2, 129.6, 129.2, 128.6, 123.1.

4.3.4. 2-Chlorophenyl benzoate (3ad)²¹. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.25–8.24 (m, 2H), 7.68–7.65 (m, 1H), 7.55–7.49 (m, 3H), 7.36–7.33 (m, 1H), 7.30–7.28 (m, 2H). ¹³C NMR

(CDCl₃, 125 MHz): δ 164.3, 147.3, 133.9, 130.4, 128.9, 128.7, 127.8, 127.1, 123.9.

4.3.5. 3-Bromophenyl benzoate (**3ae**)²². White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.19 (d, J =7.5 Hz, 2H), 7.67–7.64 (m, 1H), 7.54–7.51 (m, 2H), 7.42–7.41 (m, 2H), 7.32–7.29 (m, 1H), 7.19–7.17 (m, 1H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.7, 151.4, 133.9, 130.5, 130.2, 129.1, 128.7, 125.3, 122.5, 120.6.

4.3.6. 4-Cyanophenyl benzoate (**3af**)²³. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.12 (d, J =8.0 Hz, 2H), 7.67 (d, J =8.2 Hz, 2H), 7.60–7.59 (m, 1H), 7.48–7.45 (m, 2H), 7.30 (d, J =8.2 Hz, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.3, 154.3, 134.2, 133.7, 130.3, 128.8, 128.7, 122.9, 118.2, 109.9.

4.3.7. Phenyl benzoate (**3ag**)^{14a}. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.21 (d, J =7.5 Hz, 2H), 7.66–7.63 (m, 1H), 7.53–7.50 (m, 2H), 7.45–7.42 (m, 2H), 7.29–7.21 (m, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.2, 151.0, 133.6, 130.2, 129.6, 129.5, 128.6, 125.9, 121.7.

4.3.8. *p*-Tolyl benzoate (**3ah**)^{14a}. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.20 (d, J =7.0 Hz, 2H), 7.63–7.49 (m, 3H), 7.25–7.09 (m, 4H), 2.37 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.4, 148.7, 135.5, 133.5, 130.2, 130.0, 129.7, 128.6, 121.4, 20.9.

4.3.9. 4-Methoxyphenyl benzoate (**3ai**)²². White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.14–8.12 (m, 2H), 7.57–7.54 (m, 1H), 7.45–7.42 (m, 2H), 7.06 (d, J =9.0 Hz, 2H), 6.87 (d, J =9.0 Hz, 2H), 3.75 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.5, 157.3, 144.4, 133.5, 130.1, 129.6, 128.5, 122.4, 114.5, 55.6.

4.3.10. 3-Methoxyphenyl benzoate (**3aj**)²⁴. Pale yellow liquid. ¹H NMR (CDCl₃, 500 MHz): δ 8.14–8.12 (m, 2H), 7.59–7.56 (m, 1H), 7.46–7.43 (m, 2H), 7.27–7.24 (m, 1H), 6.77–6.70 (m, 3H), 3.75 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.1, 160.5, 151.9, 133.6, 130.2, 129.9, 129.6, 128.5, 113.9, 111.9, 107.7, 55.5.

4.3.11. 2-Methoxyphenyl benzoate (**3ak**)²². White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.22 (d, J =7.5 Hz, 2H), 7.64–7.61 (m, 1H), 7.52–7.49 (m, 2H), 7.25–7.23 (m, 1H), 7.16–7.15 (m, 1H), 7.02–6.98 (m, 2H), 3.82 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.8, 151.4, 140.0, 133.4, 130.3, 129.5, 128.5, 126.9, 122.9, 120.8, 112.6, 55.9.

4.3.12. Naphthalen-2-yl benzoate (**3al**)²⁰. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.26 (d, J =7.5 Hz, 2H), 7.90–7.82 (m, 3H), 7.70–7.64 (m, 2H), 7.54–7.46 (m, 4H), 7.37–7.35 (m, 1H). ¹³C NMR (CDCl₃, 125 MHz): δ 165.4, 148.7, 133.9, 133.7, 131.6, 130.2, 129.6, 129.5, 128.6, 127.8, 127.7, 126.6, 125.8, 121.3, 118.7.

4.3.13. 4-Nitrophenyl 4-methylbenzoate (**3ba**)²⁵. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.26–8.24 (m, 2H), 8.01 (d, J =8.0 Hz, 2H), 7.35–7.33 (m, 2H), 7.27 (d, J =8.0 Hz, 2H), 2.40 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.3, 155.9, 145.4, 145.3, 130.4, 129.5, 125.8, 125.3, 122.7, 21.8.

4.3.14. 4-Nitrophenyl 2-methylbenzoate (**3ca**)²⁶. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.33 (d, J =9.0 Hz, 2H), 8.18–8.16 (m, 1H), 7.54–7.51 (m, 1H), 7.42–7.40 (m, 2H), 7.37–7.34 (m, 2H), 2.68 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.6, 155.8, 145.4, 141.9, 133.4, 132.2, 131.3, 127.4, 126.1, 125.3, 122.7, 22.0.

4.3.15. 4-Nitrophenyl 4-ethylbenzoate (**3da**)²⁷. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.33 (d, J =9.0 Hz, 2H), 8.12 (d, J =8.5 Hz, 2H), 7.42 (d, J =9.0 Hz, 2H), 7.37 (d, J =8.0 Hz, 2H), 2.77 (q, J =7.5 Hz,

2H), 1.30 (t, J =7.5 Hz, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.3, 155.9, 151.5, 145.3, 130.5, 128.3, 126.0, 125.2, 122.7, 29.1, 15.2.

4.3.16. 4-Nitrophenyl 4-methoxybenzoate (**3ea**)²⁵. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.25 (d, J =9.0 Hz, 2H), 8.08 (d, J =9.0 Hz, 2H), 7.33 (d, J =9.0 Hz, 2H), 6.94 (d, J =9.0 Hz, 2H), 3.84 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.4, 163.9, 156.0, 145.3, 132.5, 125.2, 122.7, 120.7, 114.1, 55.6.

4.3.17. 4-Nitrophenyl 1-naphthoate (**3fa**)²⁸. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 9.01 (d, J =8.5 Hz, 1H), 8.50–8.35 (m, 3H), 8.21–8.15 (m, 1H), 7.96–7.94 (m, 1H), 7.70–7.47 (m, 5H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.6, 155.8, 145.5, 135.2, 134.0, 131.8, 131.7, 128.9, 128.6, 126.7, 125.5, 125.3, 124.6, 124.5, 122.8.

4.3.18. 4-Nitrophenyl 4-fluorobenzoate (**3ga**)²⁵. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.27–8.25 (m, 2H), 8.17–8.14 (m, 2H), 7.35–7.33 (m, 2H), 7.16–7.13 (m, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 166.5 (d, J_{C-F} =255.0 Hz), 163.3155.6, 145.5, 133.0 (d, J_{C-F} =9.5 Hz), 125.3, 124.8 (d, J_{C-F} =3.0 Hz), 122.6, 116.1 (d, J_{C-F} =22.0 Hz).

4.3.19. 4-Nitrophenyl 4-chlorobenzoate (**3ha**)²⁵. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.33 (d, J =9.0 Hz, 2H), 8.14 (d, J =9.0 Hz, 2H), 7.52 (d, J =9.0 Hz, 2H), 7.42 (d, J =9.0 Hz, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 163.4, 155.4, 145.5, 140.8, 131.6, 129.2, 126.9, 125.3, 122.5.

4.3.20. Phenyl 4-methoxybenzoate (**3eg**)^{14a}. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.08 (d, J =8.8 Hz, 2H), 7.36–7.33 (m, 2H), 7.20–7.17 (m, 1H), 7.13–7.12 (m, 2H), 6.90 (d, J =8.8 Hz, 2H), 3.82 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.9, 163.9, 151.1, 132.2, 129.4, 125.7, 121.9, 121.8, 113.8, 55.5.

4.3.21. Phenyl 4-nitrobenzoate (**3ig**)^{14g}. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.33–8.28 (m, 4H), 7.41–7.38 (m, 2H), 7.26–7.23 (m, 1H), 7.18–7.16 (m, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 163.3, 150.9, 150.5, 135.0, 131.3, 129.7, 126.4, 123.7, 121.4.

4.3.22. Phenyl 3-nitrobenzoate (**3jg**)^{14a}. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 9.04 (s, 1H), 8.54–8.49 (m, 2H), 7.38 (t, J =8.0 Hz, 1H), 7.48–7.45 (m, 2H), 7.33–7.30 (m, 1H), 7.26–7.23 (m, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 163.1, 150.5, 148.4, 135.8, 131.4, 129.9, 129.7, 128.0, 126.4, 125.1, 121.4.

4.3.23. Phenyl 4-(trifluoromethyl)benzoate (**3kg**)²⁹. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.33 (d, J =8.0 Hz, 2H), 7.79 (d, J =8.5 Hz, 2H), 7.45 (t, J =8.0 Hz, 2H), 7.30 (t, J =7.5 Hz, 1H), 7.23 (d, J =8.0 Hz, 2H). ¹³C NMR (CDCl₃, 125 MHz): δ 164.0, 150.7, 135.0 (d, J_{C-F} =32.5 Hz), 132.9, 130.6, 129.6, 126.2, 125.6 (d, J_{C-F} =3.8 Hz), 123.1 (d, J_{C-F} =271.2 Hz), 121.5.

4.3.24. Methyl phenyl terephthalate (**3lg**)^{14g}. White solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.27 (d, J =8.5 Hz, 2H), 8.16 (d, J =8.5 Hz, 2H), 7.46–7.43 (m, 2H), 7.29 (t, J =7.5 Hz, 1H), 7.23 (d, J =7.6 Hz, 2H), 3.98 (s, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 166.2, 164.4, 150.8, 134.5, 133.4, 130.1, 129.7, 129.6, 126.1, 121.6, 52.5.

4.3.25. Phenyl 4-cyanobenzoate (**3mg**)^{14h}. Pale yellow solid. ¹H NMR (CDCl₃, 500 MHz): δ 8.31 (d, J =8.0 Hz, 2H), 7.82 (d, J =8.0 Hz, 2H), 7.47–7.44 (m, 2H), 7.32–7.21 (m, 3H). ¹³C NMR (CDCl₃, 125 MHz): δ 163.6, 150.6, 133.5, 132.4, 130.6, 129.7, 126.4, 121.4, 117.8, 117.0.

4.3.26. 4-Nitrophenyl propionate (**3na**)³⁰. Pale yellow liquid. ¹H NMR (CDCl₃, 500 MHz): δ 8.27 (d, J =7.0 Hz, 2H), 7.29 (d, J =7.0 Hz,

2H), 2.64 (q, $J=7.5$ Hz, 2H), 1.28 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (CDCl_3 , 125 MHz): δ 172.0, 155.5, 145.3, 125.2, 122.4, 27.7, 8.9.

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Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.tet.2011.06.060.

References and notes

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